Hints: Methods of proof – 3. The extremal principle and compactness

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. (Sylvester–Gallai, Kelly's proof.) Consider all pairs (P, ℓ), where ℓ is a line through at least two of the points and P is one of the points not on ℓ. There are finitely many such pairs and at least one (the points are not collinear). Choose the pair with the smallest distance $d(P, \ell) > 0$.

Hint 2. Suppose ℓ contains three of the points. Drop the perpendicular PF to ℓ. Two of the three points, say B and C, lie on the same (closed) side of F, with B between F and C (possibly $B = F$).

Hint 3. Then the distance from B to the line PC is smaller than $d(P, \ell)$ (compare the right triangle PFC with the smaller similar picture at B), so the pair (B, PC) beats (P, ℓ). Hence ℓ contains exactly two points.

Hint 4. (The lens.) Let A, B be a closest pair. The lens is the intersection of the two discs of radius $|AB|$ centred at A and B; a point C inside it would have $|CA| < |AB|$ (or $|CB| < |AB|$), contradicting the choice. Note that here the extremal choice is the whole proof.

Problem 2

Hint 1. For the first claim: a continuous f on a compact K is bounded (otherwise take x_k with $|f(x_k)| > k$ and a convergent subsequence). Let $M = \sup f$, pick x_k with $f(x_k) \to M$, pass to a convergent subsequence $x_k \to x^* \in K$ and use continuity.

Hint 2. The unit sphere is compact and $q(x) = \langle Ax, x\rangle$ is continuous, so q attains its maximum λ at some unit x_0. For $y \perp x_0$ consider $x(t) = (x_0 + ty)/|x_0 + ty|$; the function $t \mapsto q(x(t))$ has a maximum at $t = 0$, and its derivative there is $2\langle Ax_0, y\rangle$.

Hint 3. Hence $Ax_0 \perp x_0^{\perp}$, i.e. $Ax_0 = \lambda x_0$ with $\lambda = q(x_0)$. Every eigenvalue μ with unit eigenvector v satisfies $\mu = q(v) \leqslant \lambda$, so λ is the largest one.

Hint 4. For the eigenbasis: $W = x_0^{\perp}$ is A-invariant, since $\langle Aw, x_0\rangle = \langle w, Ax_0\rangle = \lambda\langle w, x_0\rangle = 0$, and $A|_W$ is again symmetric. Apply the induction hypothesis to W and add x_0.

Problem 3

Hint 1. Parametrise ellipsoids as $E = \{Ax + c : \ |x| \leqslant 1\}$ with A symmetric positive semidefinite; $\operatorname{vol} E = \det A \cdot \operatorname{vol} B^n$. Show that the set of (A, c) with $E \subset K$ is closed and bounded (since K is bounded), so the continuous function $\det A$ attains its maximum; it is positive because K has interior points.

Hint 2. Uniqueness of the shape: if (A_1, c_1) and (A_2, c_2) are both maximal, then $(\frac{A_1+A_2}{2}, \frac{c_1+c_2}{2})$ also gives an ellipsoid in K (each of its points is the midpoint of a point of E_1 and a point of E_2). Use $\det \frac{A_1+A_2}{2} \geqslant \sqrt{\det A_1 \det A_2}$, with equality only if $A_1 = A_2$ (diagonalise $A_1^{-1/2} A_2 A_1^{-1/2}$ and use AM–GM on its eigenvalues).

Hint 3. Uniqueness of the centre: if $A_1 = A_2$ but $c_1 \neq c_2$, make $A_1 = I$ by an affine map. Then K contains the convex hull of two unit balls with centres at distance $d > 0$, and this hull contains

the ellipsoid centred at the midpoint with semi-axis $1 + d/2$ along $c_2 - c_1$ and 1 in the orthogonal directions: a larger volume.

Hint 4. The contact points (John's theorem): if the maximal ellipsoid is the unit ball B, then there are points $u_1, \dots, u_m \in \partial K \cap S^{n-1}$ and weights $c_i > 0$ with $\sum c_i u_i u_i^T = I_n$ and $\sum c_i u_i = 0$. Consequently the contact points are not contained in any closed half-sphere and $K \subset nB$ ($K \subset \sqrt{n}B$ if K is symmetric).

Problem 4

Hint 1. Let $S = \{x \in [0,1] : f(x) \geqslant x\}$; $0 \in S$, so $s = \sup S$ exists. Rule out $f(s) > s$ and $f(s) < s$ separately using continuity at s.

Hint 2. If $f(s) > s$, then $s < 1$ and $f(x) > x$ on a right neighbourhood of s, contradicting $s = \sup S$. If $f(s) < s$, then $f(x) < x$ on some $(s - \delta, s]$, so $S \subset [0, s - \delta]$, again a contradiction. Hence $f(s) = s$.

Hint 3. Counterexamples: $f = 1$ on $[0, \frac{1}{2})$, $f = 0$ on $[\frac{1}{2}, 1]$ has no fixed point (the step "$f(x) > x$ near s" fails); $f(x) = x/2$ on $(0, 1)$ has none: there the set S is empty, because the starting point $0 \in S$ is missing, and the would-be fixed point 0 lies outside the open interval, which is not compact.

Problem 5

Hint 1. If $\bigcap F_i = \varnothing$, the complements $X \setminus F_i$ form an open cover of X. Take a finite subcover and see what it says about a finite intersection.

Hint 2. For the graph $G = (V, E)$ let $X = \{1, \dots, k\}^V$ with the product topology; it is compact by Tychonoff (for countable V a diagonal argument suffices). For each edge uv let $C_{uv} = \{c \in X : c(u) \neq c(v)\}$; it depends on two coordinates only, hence is closed.

Hint 3. A finite family $C_{u_1 v_1}, \dots, C_{u_m v_m}$ has a common point because the finite subgraph on these edges is k-colourable (extend the colouring arbitrarily). By the first part there is $c \in \bigcap_{uv \in E} C_{uv}$: a proper colouring of G.

Hint 4. In the finite case one would simply take a colouring of the whole graph (or an extremal partial colouring and extend it). In the infinite case there is no "largest finite subgraph"; the common point of the closed sets, i.e. a consistent limit of colourings of larger and larger finite subgraphs, is exactly what compactness supplies in place of the extremal object.